```
  node ←
treap.root
          #2
  parent ← null
  newNode ← new Node(key, priority)
                            #3
  while node != null do
                  #4
    parent ←
node                                                          #5
    if node.key <= key
then                                                         #6
      node ← node.left
    else

  #7
      node ← node.right
  if parent == null then
                            #8
    treap.root ← newNode
                                          #9
    return
  elsif parent.key <= key
then                                              #10
    parent.left ← newNode
  else

    parent.right ← newNode
  newNode.parent ←
parent                                                    #11
  while newNode.parent != null
      and newNode.priority > newNode.parent.priority do
                  #12
    if newNode == newNode.parent.left
then                                       #13
      rightRotate(newNode)
    else
      leftRotate(newNode)
                                    #14
  if newNode.parent == null
then                                                #15
    treap.root ← parent
```